

InfectioGIT

Exploring FAIRness and Reusability of
Infectious Disease Computational Models
from Public Repositories

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1 Context and Objectives

Infectious diseases remain a major global public health challenge, driven by factors such as climate change, increased human mobility, and ecological disruption. Addressing these threats increasingly relies on computational modelling approaches capable of representing the complexity of host–pathogen interactions and predicting disease dynamics. Mechanistic mathematical models have therefore become essential tools for understanding infection processes, evaluating interventions, and supporting public health decision-making. In this context, infectious disease digital twins — virtual *in silico* representations of biological systems integrating mechanistic models and multi-scale data — are emerging as promising frameworks for predictive medicine and epidemic preparedness. However, their development is currently limited by the fragmentation of computational resources across heterogeneous repositories, inconsistent metadata annotation, and poor interoperability between model formats. Existing models are often difficult to locate, reuse, and integrate into larger modelling frameworks. The *InfectioGIT* project was developed to address these limitations by creating a centralized, FAIR-compliant repository of infectious disease computational models within the EUR UNITEID and One Health frameworks. The project focuses on diseases of both epidemiological and modelling relevance, including West Nile virus, dengue, chikungunya, Mpox, avian influenza, and SARS-CoV-2. Models were retrieved primarily from BioModels and complemented with resources from NCBI and Zenodo using an automated Python-based pipeline integrating model retrieval, filtering, metadata enrichment, and standardized JSON annotation aligned with COMBINE Archive standards. Beyond repository construction, *InfectioGIT* establishes a scalable and reproducible curation workflow designed to facilitate model interoperability, reuse, and future integration into infectious disease digital twin architectures.

1.1 Project Objectives

The primary objectives of the *InfectioGIT* project are:

1. **Build a centralized, FAIR-compliant repository** of infectious disease computational models, consolidating resources from heterogeneous public sources (BioModels, NCBI, Zenodo) into a unified, machine-readable knowledge base aligned with COMBINE Archive metadata standards.
2. **Assess the reusability and interoperability** of existing infectious disease computational models by analyzing their curation status, format diversity, metadata completeness, and cross-source availability across the six-disease panel.
3. **Establish a scalable, automated curation pipeline** that integrates model retrieval, filtering, quality control, and metadata enrichment, enabling rapid, reproducible repository updates as new models emerge.
4. **Identify knowledge gaps and structural barriers** to model availability and reuse, quantifying disparities between epidemiological importance and computational coverage, and proposing targeted solutions for emerging pathogens.

2 Materials and Methods

2.1 Disease Selection and Curation Criteria

Disease selection was guided by two complementary criteria: public health priority (emerging/re-emerging pathogens with climate or spillover drivers, referenced against WHO priority lists and EUR UNITEID frameworks) and computational modelling landscape (diseases with rich model histories serving as benchmarks). At the national level, three arboviruses were selected: West Nile virus, dengue, and chikungunya. At the international level: Mpox, avian influenza, and SARS-CoV-2 (as a benchmarking disease given exceptional model volume).

2.2 Model Retrieval Pipeline

For each disease, standardised keyword lists (synonyms, pathogen names, acronyms) were compiled and used to query three complementary public sources: BioModels (primary curated repository, SBML/SBML-qual via REST API), NCBI/PubMed Central (open-access literature and embedded supplementary materials), and Zenodo (multidisciplinary open-access repository for datasets and software).

2.2.1 BioModels

The `Bioservices` library provides a high-level interface to the BioModels REST API; disease-specific keyword queries retrieved matching model identifiers and metadata. The `biomodels` Python library downloads associated files and sorts them into model files (`.sbml`, `.xml`, `.omex`, `.m`, `.py`, `.java`, etc.) and metadata files (`.json`, `.csv`, images, documentation). Filtering retained only models with curation status “CURATED” and publication year ≥ 2015 . The JSON metadata files originally provided by BioModels were found to be systematically incomplete — frequently lacking model summaries, ontology annotations, organism references, or publication details. To ensure a uniform and queryable representation across the repository, the `metadata.json` files were therefore not retained as-is but fully regenerated from scratch for every BioModels entry. The regenerated files follow the unified *InfectioGIT* metadata schema and consolidate information drawn from the BioModels API, the associated SBML annotations, and external ontology sources.

2.2.2 NCBI PubMed Central

A systematic mining pipeline retrieves supplementary materials, SBML models, and executable code from open-access literature, addressing the limitation that many computational resources remain embedded in published articles. The NCBI Entrez API via `Biopython` searches PubMed for articles (≥ 2015) containing supplementary materials, source code, SBML, and computational models. PMIDs are mapped to PMCIDs using the Entrez `elink` service, and open-access packages are retrieved from the PMC OA API. Downloaded archives (ZIP, TAR.GZ) are recursively extracted using `zipfile` and `tarfile`. A file classification system sorts files into data (executable code: `.py`, `.r`, `.m`, `.ipynb`; SBML models: `.sbml`, `.xml`, `.sedml`, `.omex`, `.cellml`; bioinformatics: `.vcf`, `.fasta`, `.gff`; data formats: `.json`, `.csv`, `.tsv`, `.xlsx`) and metadata (images, documentation, PDFs). A validation step assesses the executable status of Python and

R files, excluding non-executable code snippets, inline examples, and documentation. Citation-based ranking prioritises high-impact publications: citation counts are retrieved via **Biopython** Entrez, and models are ranked by citation metrics, journal quality, and publication year. Files are organised hierarchically by disease and PMC ID into **data/** and **metadata/** subdirectories with CSV and JSON manifests.

2.2.3 Zenodo

The Zenodo REST API (via **requests**) retrieves published records containing computational models. Searches target file types **.xml**, **.sbml**, **.omex**, and **.r**, filtered by publication status (published only) and sorted by recency. For each record, file metadata is stored as JSON, and model files are downloaded into disease-specific directories. SBML files and executable code are prioritised for integration.

2.2.4 Quality Control, Cleanup, and Metadata Enrichment

After retrieval from all three sources, a Python cleanup script removed file extensions not corresponding to computational models. Each file was manually verified to ensure only genuine modelling files were retained. Empty directories were automatically deleted, and a cleanup report documented files retained and removed. A unified metadata schema was designed in accordance with COMBINE Archive metadata standards and FAIR data principles, implemented in JSON for its lightweight structure, human readability, and compatibility with programmatic querying. For each model, the metadata record captures: unique model identifier, associated disease name, ontology identifiers, model type (e.g. SIR, ODE-based, logical), modelled organism, associated publication reference, author information, and publication year. An automated Python enrichment pipeline populated missing ontology annotations, ensuring uniform metadata standards across the repository regardless of original curation status.

2.3 Programming Environment

The pipeline was developed in Python 3. Key dependencies include **Bioservices** (BioModels REST API), **Biopython** (NCBI Entrez and publication metadata), **requests** (HTTP file downloads), and standard libraries for XML and JSON parsing. **libsbml-python** was used for SBML model validation and metadata schema enforcement.

2.4 Statistical Analysis and Data Quantification

Statistical analysis was performed in R using **tidyverse**, **ggplot2**, and **janitor**. For all three data sources, the following analyses were conducted: (1) publication date trends over time and by disease category, (2) distribution of modelling approaches across diseases, (3) file format inventory and density heatmaps by disease, and (4) model count summaries per category. Outputs comprise 8 PNG plots and summary statistics as CSV files.

2.5 Database Infrastructure

A SQLite relational database integrates models from all three sources, comprising eight principal tables (Model, Disease, Pathogen, Host, Tissue, CellType, Dataset) with junction tables enforcing referential integrity. Source is automatically detected from JSON metadata keys; source-specific

parsers normalize heterogeneous data structures. Parsed metadata are inserted via atomic transactions, enabling complex SQL queries for cross-source inventory analysis, temporal trends, and format comparisons. SQLite was selected for portability and minimal deployment overhead.

3 Results

3.1 BioModels Database Analysis

3.1.1 Model Inventory

BioModels search retrieved a total of 160 models distributed unevenly across six infectious disease categories. COVID-19 dominated the repository with the highest model count, followed by influenza, HIV, and tuberculosis. Dengue and Mpox showed substantially lower representation, while West Nile virus returned no results. This distribution reflects significant disparities in computational modelling coverage across pathogens.

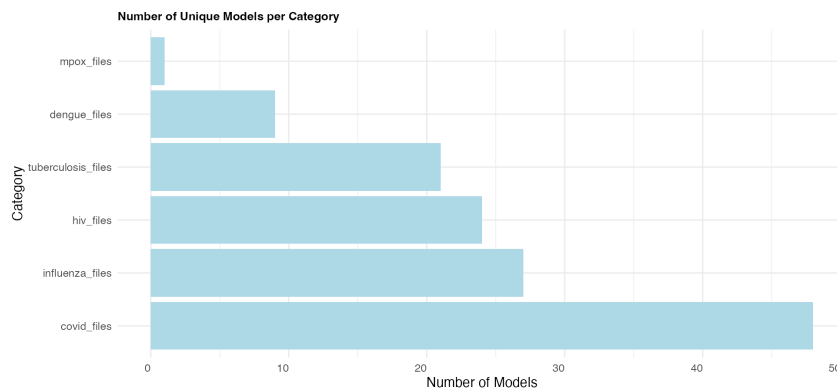


Figure 1: Number of Unique Models per Disease Category

3.1.2 Publication Timeline and Research Evolution

Analysis of publication dates spanning 1991–2026 revealed two distinct patterns. Early publications (1991–2010) showed sporadic, low-volume model development (1–7 models/year). A transition period (2011–2019) exhibited moderate but steady growth. A dramatic surge occurred in 2021, with 24 models published, followed by a decline to 8 models in 2025, reflecting the rapid modelling response to the COVID-19 pandemic. Disease-specific trends show COVID-19 publications concentrated in 2021–2023, while influenza, HIV, and tuberculosis maintained stable output over time. Models are distributed across diverse publication venues, with preprints and *Mathematical Biosciences* as the most frequent sources, followed by *Journal of Theoretical Biology*, *PLoS One*, and *Viruses* (see Appendix A.1).

3.1.3 Curation Quality and Reliability

Curation status analysis reveals substantial heterogeneity across diseases. COVID-19, influenza, HIV, and tuberculosis show mixed curation levels (approximately 45–65% curated). In contrast, **dengue and Mpox models are 100% non-curated**, indicating the absence of formal BioModels quality validation in these categories. Temporal trends demonstrate declining curation capacity relative to model production (see Appendix A.2).

3.1.4 Modelling Approaches and Mathematical Frameworks

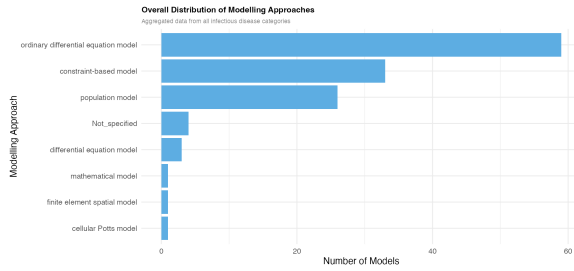


Figure 2: Overall Distribution of Modelling Approaches

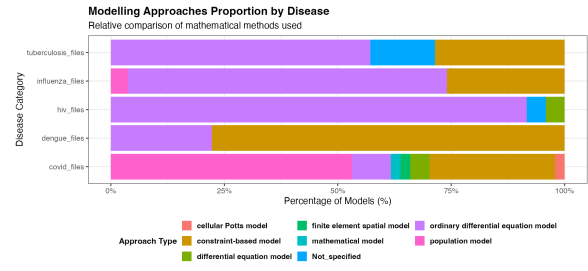


Figure 3: Modelling Approaches by Disease Category

The distribution of modelling approaches reveals distinct computational strategies aligned with pathogen characteristics and research maturity. Endemic diseases with established epidemiological foundations (HIV, tuberculosis, influenza) converge on deterministic temporal frameworks, while emerging or metabolically-focused pathogens (dengue, Mpox) demonstrate preference for constraint-based methods. COVID-19’s methodological diversity reflects both the pandemic’s multidisciplinary research response and the heterogeneous scales of analysis required. For detailed definitions of each approach, see Appendix A.3.

3.1.5 File Format and Computational Representation

A total of 619 files were recovered across all models, representing computational implementations of mathematical frameworks. SBML XML files (.xml) dominate at 198 files (32%), followed by MATLAB scripts (.m, 153 files, 25%), CellML/ODE formats (115 combined, 19%), and SEDML formats (.sedml, 63 files, 10%).

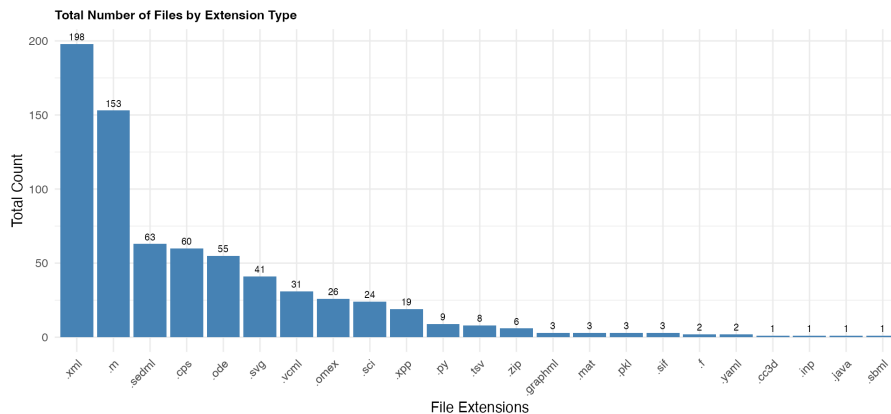


Figure 4: Total Number of Files by Extension Type Across BioModels Entries

COVID-19 models show the widest format diversity, while dengue and Mpox remain narrowly implemented. The disease-specific heatmap of format density is provided in Appendix A.4.

3.2 NCBI Article Analysis

3.2.1 File Type Distribution in NCBI Downloads

Analysis of NCBI article downloads across eight infectious disease categories (chikungunya, COVID-19, dengue, HIV, influenza, Mpox, tuberculosis, and West Nile virus) reveals a strong predominance of metadata over computational models. A total of 1,247 files were classified across data and metadata categories. File type distribution demonstrates that NCBI downloads are overwhelmingly composed of metadata files (PDFs, images, documents), while computational model files remain scarce. This highlights a structural limitation: although NCBI hosts extensive infectious disease literature, it rarely provides reusable computational models. Detailed distributions are provided in Appendix A.5.

3.2.2 SBML Models in Article PMC6620235

A notable exception is article PMC6620235, which contains 20 SBML files representing 10 unique organisms. This demonstrates that computational models can be embedded within single publications and may span multiple biological systems beyond the nominal disease classification. The article includes influenza-related organisms as well as cross-pathogen models including tuberculosis, *Klebsiella pneumoniae*, and *Staphylococcus aureus*, illustrating the multi-organism nature of systems biology modelling.

Table 1: SBML Organisms in PMC6620235

No.	Organism/Model	Type
1	<i>Haemophilus influenzae</i>	Influenza
2	integrated <i>M. tuberculosis</i>	Tuberculosis
3	<i>Klebsiella pneumoniae</i>	Other
4	<i>Pseudomonas aeruginosa</i>	Other
5	<i>Pseudomonas putida</i>	Other
6	<i>Staphylococcus aureus</i>	Other

3.3 Zenodo Database Analysis

3.3.1 Model Retrieval and Valid Model Count

The Zenodo search pipeline retrieved computational models and analysis scripts across the infectious disease panel. After applying the two-stage filtering process (keyword search followed by modelling-file detection), the pipeline identified records with sufficient modelling-related content. Manual verification was subsequently applied to determine whether retrieved files corresponded to executable computational models or instead represented epidemiological summaries, datasets, documentation files, or auxiliary analysis scripts. After manually checking each retrieved file, only a limited subset qualified as valid computational models. The distribution of validated models across infectious disease categories is shown in Figure 5.

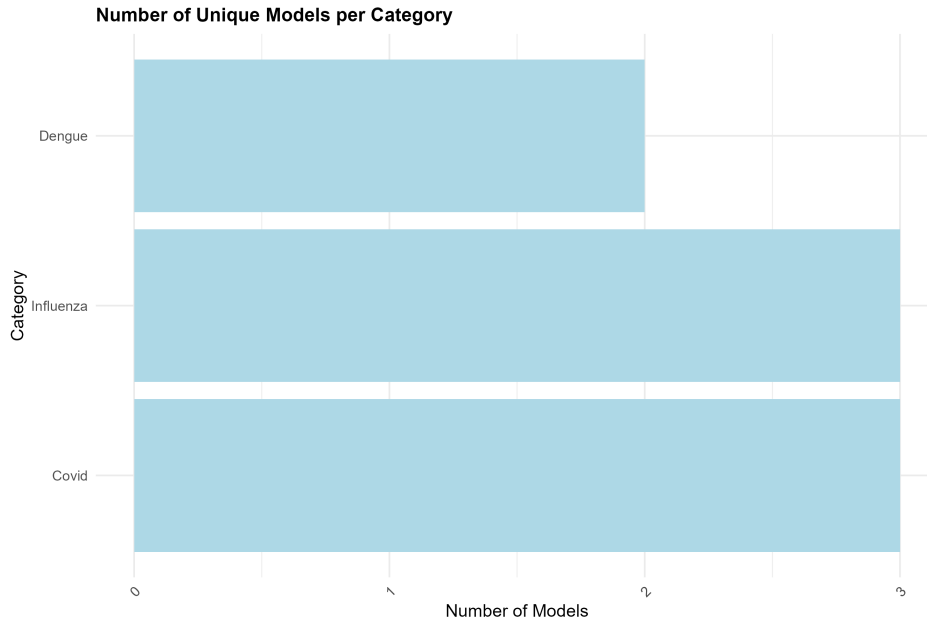


Figure 5: Number of Unique Valid Computational Models Retrieved from Zenodo Across Infectious Disease Categories

A total of 8 valid computational models were identified across Zenodo, contrasting sharply with the 160 models retrieved from BioModels — a retrieval rate approximately 20-fold lower.

4 Summary, Assessment and Perspectives

4.1 Assessment of Results Obtained

The *InfectioGIT* pipeline successfully assembled and integrated 170 computational models from three major public repositories: 160 from BioModels, 8 from Zenodo, and 2 from NCBI/PMC. The analysis revealed stark disparities in model availability: COVID-19 models dominate BioModels (81 models), while West Nile virus is entirely absent and Mpox shows minimal representation (3 models). File format analysis identified SBML XML as the predominant representation (32%), followed by MATLAB scripts (25%), reflecting the divide between formal model standards and legacy implementation formats. Critically, BioModels curation rates decline dramatically for emerging diseases: dengue and Mpox models show 100% non-curation status, indicating absent quality assurance.

The metadata enrichment pipeline successfully normalized heterogeneous JSON structures from all three sources into a unified COMBINE Archive-compliant schema, enabling cross-source querying and FAIR-compliant resource discovery. The SQLite database supports complex analytical queries, facilitating temporal trend analysis (publication surge in 2021–2023 COVID-19 response, stable output for endemic diseases) and disease-specific format inventories.

4.2 Assessment of Objectives and Deliverables

Objective 1: Build a centralized, FAIR-compliant repository. Successfully achieved. A unified repository integrating 170 computational models was constructed with standardized

JSON metadata aligned with COMBINE Archive standards and a SQLite backend enabling FAIR-compliant querying and interoperability.

Objective 2: Assess reusability and interoperability. Successfully achieved. Comprehensive analysis quantified barriers to reuse: BioModels dominance, low curation for emerging pathogens, format fragmentation, and absence of executable models in supplementary materials. These findings directly address the project’s diagnostic goal.

Objective 3: Establish a scalable curation pipeline. Substantially achieved. A fully automated Python pipeline integrates retrieval from three sources, quality control, metadata enrichment, and database ingestion. However, manual file verification and cross-source deduplication remain necessary, preventing complete end-to-end automation.

Objective 4: Identify knowledge gaps and propose solutions. Successfully achieved. The report documents specific scarcity gaps (West Nile virus absence, Mpox underrepresentation) and proposes concrete, evidence-based solutions (FAIR enforcement at publication stage, GitHub mining, semi-automated curation pipelines, mechanism-oriented indexing, community training).

4.3 Limitations and Future Directions

Several limitations warrant acknowledgment:

- **Full automation:** Manual file verification and cross-source deduplication remain necessary. Future work should implement machine learning classifiers for automated file validation.
- **GitHub mining:** Systematic extraction from GitHub repositories, identified as a high-priority gap, was not implemented due to time constraints.
- **Model validation:** The pipeline catalogues models but does not execute or validate computational correctness. Comprehensive validation would require additional infrastructure.
- **Longitudinal tracking:** The repository captures a static snapshot (January–May 2026). Ongoing monitoring of deposition rates and curation trends should be established.

Future directions include extending model retrieval to GitHub and GitLab platforms, implementing semi-automated curation pipelines with expert review, systematic mining of non-English literature, and development of mechanism-oriented indexing alongside disease-centric organization. Additionally, federated indexing across BioModels, NCBI, Zenodo, GitHub, and domain-specific databases would improve discoverability without sacrificing curation quality.

5 Project Management

5.1 Project Timeline

A Gantt diagram detailing the project timeline and major milestones is presented in Figure 6.



Figure 6: Project Gantt Diagram — Timeline of major tasks and milestones (January—May 2026)

5.2 Task Distribution and Contributions

The project was organized around four primary work streams: (1) model retrieval from BioModels, (2) NCBI/PubMed Central mining, (3) Zenodo database integration, and (4) database infrastructure and statistical analysis.

5.3 Repository Structure

The project code and documentation are organized in a version-controlled GitHub repository: <https://github.com/thalieh/InfectioGIT>

The repository structure is organized as follows:

```

InfectioGIT/
├── src/
│   ├── Biomodels/
│   ├── NCBI/
│   ├── Zenodo/
│   └── Database/
├── Results/
│   ├── Models/ (by disease)
│   └── Statistics/
├── docker/
├── project-report/
├── README.md
└── LICENCE

```

6 Use of Generative AI Tools

In accordance with the project guidelines, we disclose the use of generative AI tools as follows:

6.1 Tools and Applications

Three generative AI platforms were used throughout the project:

- **GPT-4o** (OpenAI)
- **Gemini 2.0** (Google DeepMind)
- **ChatGPT** (OpenAI)

6.2 Scope and Usage

Generative AI was used in the following contexts:

- **Code development and optimization:** GPT-4o and Gemini 2.0 were employed to review, debug, and refactor Python code for the retrieval pipeline, database integration, and statistical analysis scripts. These tools assisted in identifying inefficiencies, suggesting algorithmic improvements, and generating clear inline documentation.
- **Technical documentation:** AI tools were used to generate standardized docstrings, function annotations, and inline comments explaining complex logic, ensuring code clarity and maintainability for future developers.
- **Report writing and structure:** ChatGPT and GPT-4o were used to refine and structure sections of this report, improve sentence clarity, and ensure consistent academic tone and formatting.

6.3 Limitations and Human Oversight

All AI-generated content was reviewed, validated, and edited by the authors. AI outputs were used as drafts or suggestions only; final decisions on code logic, scientific interpretation, and manuscript content remained under direct human control. Code functionality was verified through testing, and all scientific claims were validated against primary sources and the authors' own analyses.

Author Contributions

The contributions of each author reflect both primary responsibility for specific work streams and shared involvement across the entire project:

Thalie Holmiere led the BioModels retrieval pipeline, including API integration, filtering logic, and quality control workflows. She authored the initial project specification document and was instrumental in defining the metadata schema in coordination with team members. She contributed substantially to the statistical analyses in R and participated in manuscript drafting.

Aya Sebbah developed the NCBI/PubMed Central mining pipeline, including archive extraction, file classification, citation-based ranking, and integration with the central database. She led the Docker containerization effort, ensuring reproducibility across development and deployment environments. She led the writing and revision of this report in L^AT_EX, managed manuscript organization, and contributed to all pipeline components including statistical visualizations.

Diana Bravais designed and implemented the unified SQLite relational database, including schema architecture, multi-source parsing logic, and query optimization. She developed the Zenodo retrieval pipeline and led the automated enrichment of metadata from ontology sources. She contributed to Python pipeline development and participated in statistical analyses.

Shared contributions: All three authors participated in code review and optimization using generative AI tools, contributed to the Python codebase across all three retrieval streams, conducted portions of the R-based statistical analyses, jointly discussed and interpreted findings, and shared equally in GitHub repository maintenance, version control, and documentation.

A Supplementary Analyses and Detailed Figures

A.1 Publication Timeline and Journal Analysis

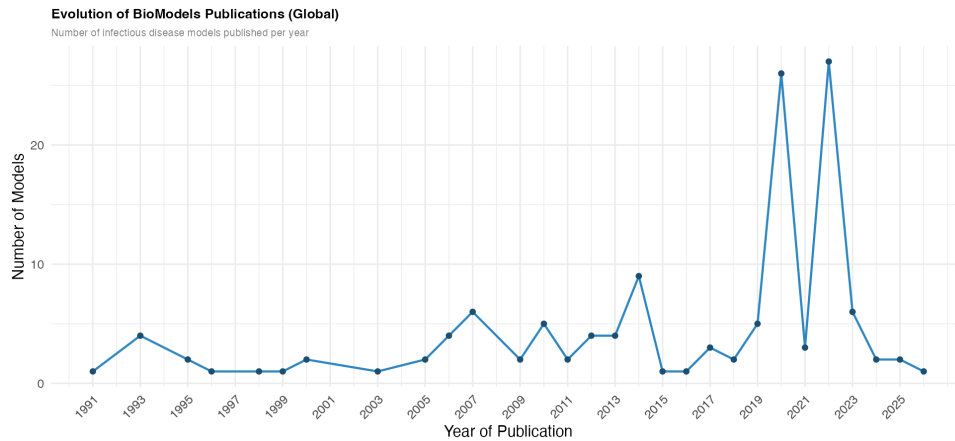


Figure 7: Evolution of BioModels Publications (Global) — Number of infectious disease models published per year (1991–2026)

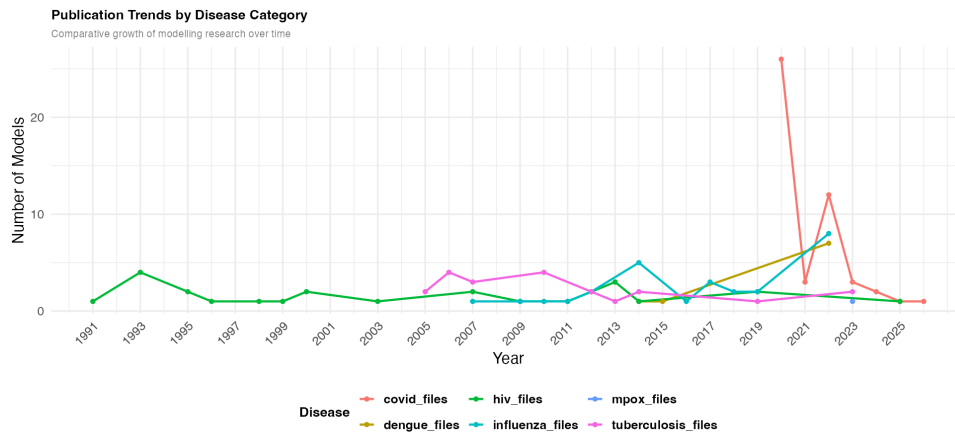


Figure 8: Publication Trends by Disease Category — Comparative growth of modelling research over time

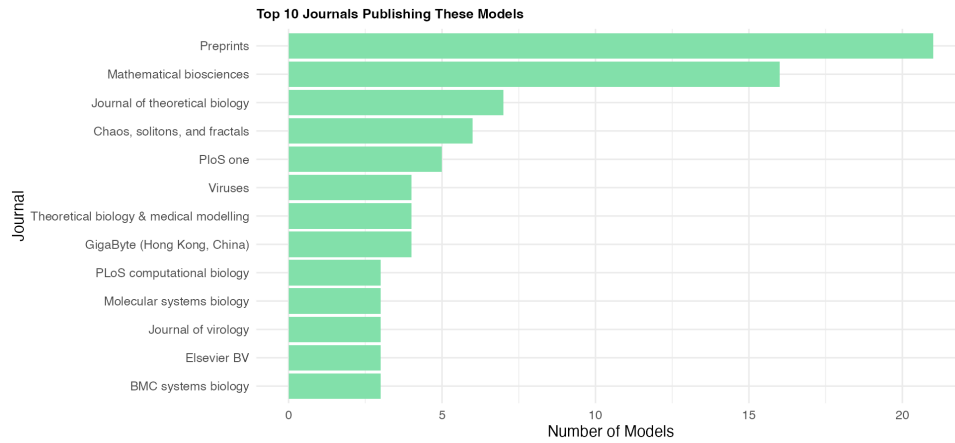


Figure 9: Top 10 Journals Publishing These Models — Most prolific publication venues for infectious disease computational models

A.2 Curation Quality Evolution

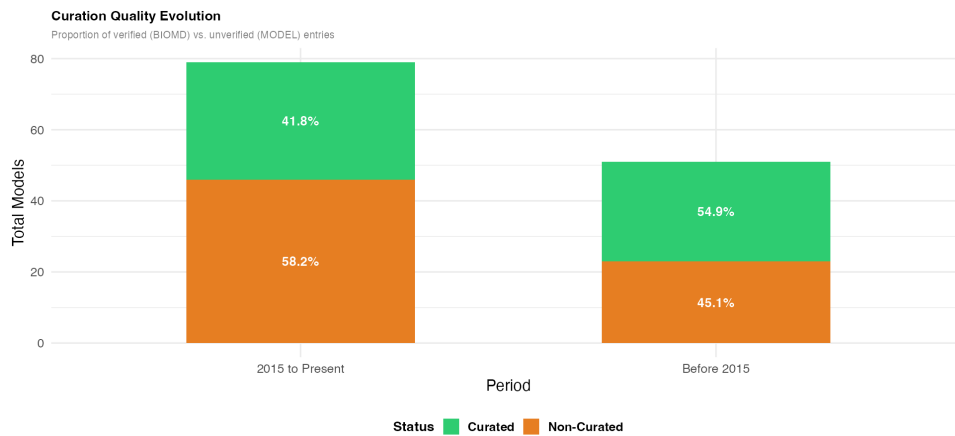


Figure 10: Curation Quality Evolution — Proportion of verified (BIOMD) vs. unverified (MODEL) entries across time periods

A.3 Modelling Approaches: Detailed Description

Table 2: Modelling Approaches Used in BioModels — Overview and descriptions of mathematical frameworks

Modelling Approach	Description
Ordinary Differential Equations (ODEs)	Deterministic continuous-time models describing temporal evolution of biological variables such as infection dynamics, immune response, or population compartments. Dominant approach across HIV, influenza, and tuberculosis.
Population-based models	Stochastic or agent-based models representing individual entities or subpopulations, allowing heterogeneity, randomness, and discrete event dynamics. Used primarily in COVID-19 and epidemiological applications.
Constraint-based models	Steady-state frameworks representing metabolic or biochemical networks under stoichiometric constraints, often applied in genome-scale metabolic reconstructions. Predominant in dengue and Mpox analyses.
Differential Equation models	Additional ODE variants and specialized differential equation formulations for capturing complex temporal dynamics.
Cellular/Potts models	Cell-based computational models representing discrete cellular entities and their interactions, used for spatial and multicellular dynamics.
Finite Element Spatial models	Spatial discretization approaches for modelling continuous processes over heterogeneous domains, relevant for tissue-level or spatial epidemic models.
Mathematical models	General mathematical frameworks not otherwise specified or hybrid approaches combining multiple methodologies.

A.4 File Format Density by Disease

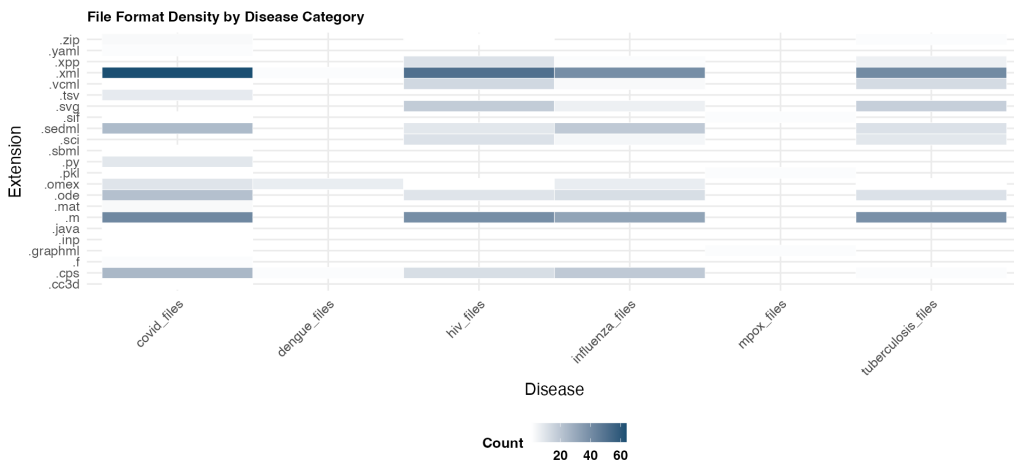


Figure 11: File Format Density by Disease Category — Heatmap showing file format distribution across diseases

A.5 NCBI Article Analysis

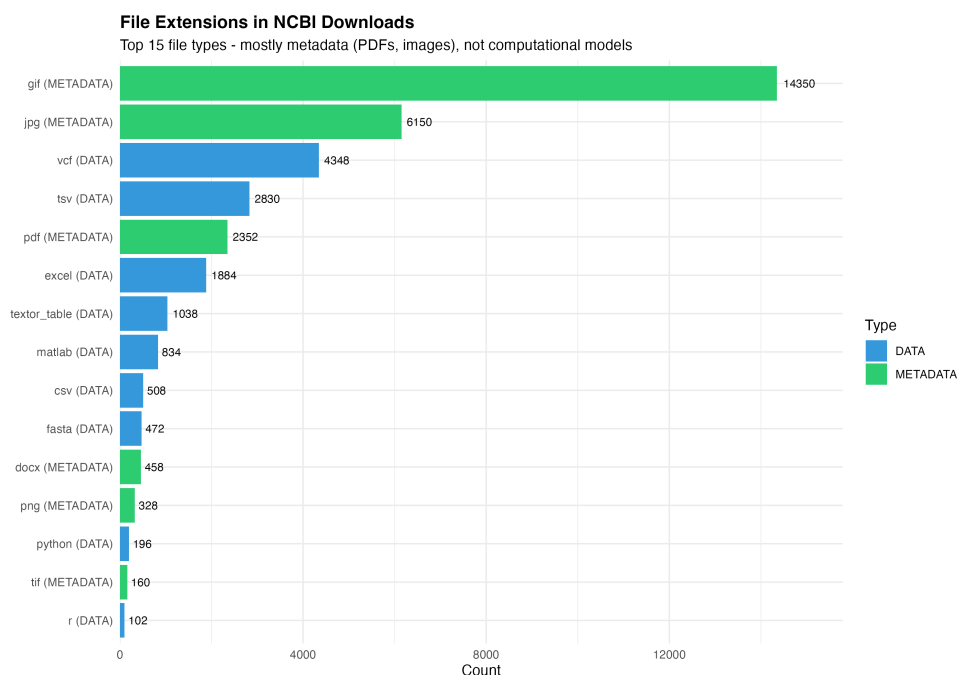


Figure 12: File Extensions Found in NCBI Downloads — Distribution of top 15 file types across disease categories, classified by type (DATA vs. METADATA)

A.6 Zenodo Detailed Analysis

A.6.1 File Extension Distribution

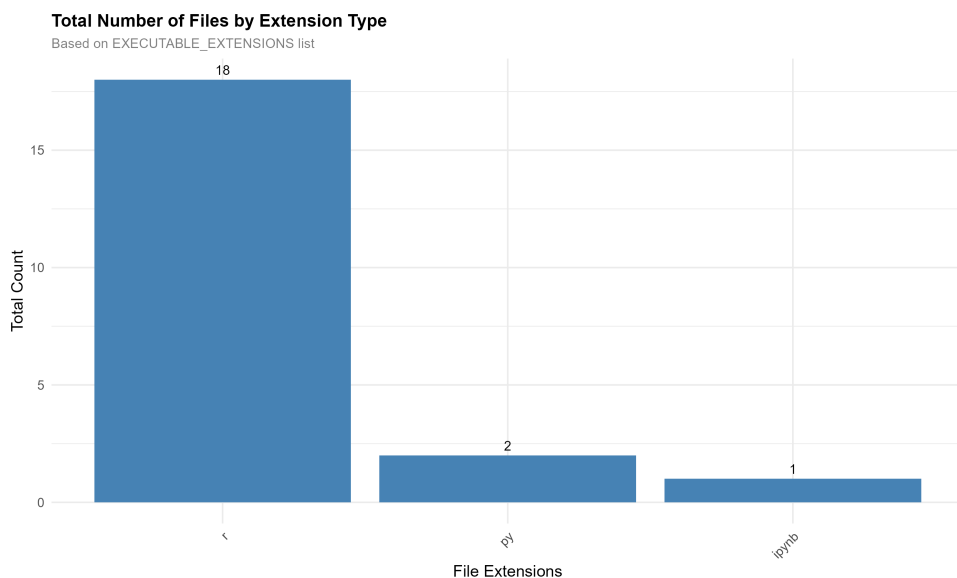


Figure 13: File Extensions of Valid Computational Models Retrieved from Zenodo — Distribution of executable file types across the 8 validated models

The distribution of executable file extensions among the validated Zenodo models is strongly skewed towards R scripts (18 files), with minimal Python representation (2 .py files and 1 .ipynb

notebook). No SBML or other standardised model files were recovered among the valid Zenodo entries, contrasting sharply with the BioModels repository where SBML XML files dominate. This pattern suggests that Zenodo’s infectious disease modelling content is primarily composed of statistical and epidemiological analysis scripts rather than mechanistic systems biology models.

A.6.2 Publication Timeline

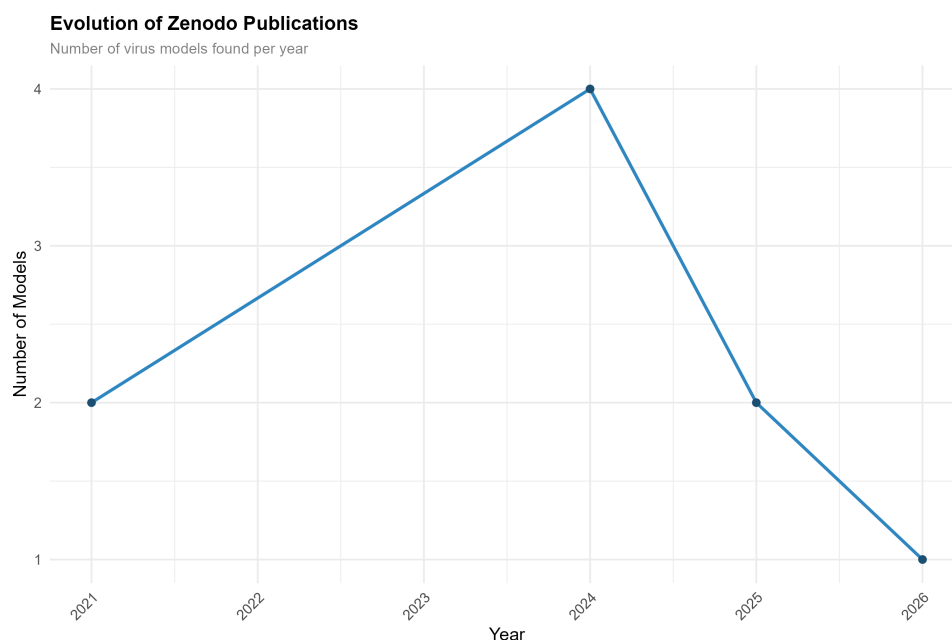


Figure 14: Evolution of Zenodo Publications — Number of valid virus models deposited per year (2021–2026)

The temporal distribution of validated Zenodo models spans a narrow window from 2021 to 2026, with deposits increasing from 2 models in 2021 to a peak of 4 models in 2024, followed by a decline to 2 models in 2025 and 1 in 2026. This recent-year skew is consistent with Zenodo’s relatively young role as a deposition platform for infectious disease modelling outputs, and contrasts with BioModels’ broader historical coverage extending back to 1991. The low absolute numbers across all years reinforce the conclusion that Zenodo currently functions as a supplementary rather than primary repository for reusable computational models in this domain.

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